

Analysis of gene origin in the first adult returns to the Cultus sockeye salmon captive breeding program

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Abstract Rigorous evaluation of the utility of captive breeding for the restoration of depleted wild salmonid fish populations has not been undertaken. In particular, little is known about the reproductive success of captively-bred individuals that are released back into an extant population and their capacity to assist in long-term population persistence. For the endangered Cultus Lake sockeye salmon population, we examined the potential genetic contribution of the first juvenile fish released from a captive breeding program upon their maturity in the natural Cultus Lake environment. Genetic analysis of 792 Cultus sockeye salmon that were spawned in captivity in 2004 and their adult progeny of 2007 and 2008 revealed a genetic bottleneck originating from 20 wild sockeye salmon hatchery-spawned at Cultus Lake in the previous generation. Pedigree analysis revealed that six of the 20 sockeye salmon spawned in 2001 (grandparents) gave rise to a majority of the hatchery spawners in 2004 (parents) and provided more than 30% of the genes in the progeny that survived to maturity in the wild. Allele frequencies and genetic diversity of the age three progeny that returned to Cultus Lake from their marine migration in 2007 reflected the bottleneck, but its genetic signature was faint among the

more genetically diverse age four fish that returned in 2008. Two-generation analysis of gene origin among fish resulting from 2004 hatchery production indicated that they contained the genetic diversity expected from 36 effective ancestors.

Keywords Sockeye salmon · Captive breeding · Genetic diversity · Endangered · Pedigree

Introduction

Populations greatly reduced in size during extended periods of low survival lose genetic diversity due to genetic drift and inbreeding, processes that may lead to reduced fitness, further reductions in survival and the possibility of an extinction vortex (Gilpin and Soulé 1986). Hatchery programs are in widespread use for Pacific salmonid populations but little assessment of their effectiveness for wild population recovery has been conducted (Waples et al. 2007). A fundamental question remains as to whether the immediate increase in abundance provided by hatchery production can offset the expected reduction in fitness for re-introduced hatchery individuals that breed in the wild (Araki et al. 2008; Fraser 2008). Captive breeding, in which individuals are removed from the natural environment for at least one entire life cycle to increase survival and reproductive capacity, can rapidly increase juvenile production in highly fecund species such as sockeye salmon. Pedigreed breeding programs can minimize the loss of genetic diversity due to inbreeding and random processes (e.g. O'Reilly and Harvie 2010), but pose their own challenges to maintenance of fitness in the wild environment (Fraser 2008). The alteration or relaxation of natural selection experienced in captivity may engender genetic

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alterations that impose a cost to fitness upon those individuals ultimately re-introduced to the natural environment (Lynch and O'Hely 2001; Araki et al. 2008).

In this case study, we evaluate some genetic consequences of a program of captive breeding designed to recover an endangered sockeye salmon population in Cultus Lake, located in southern British Columbia (BC). The Cultus Lake population is a genetically distinctive member of the 'late run' group of Fraser River sockeye populations that are so-named because of their adult time of entry into the Fraser River (historically, mid September through October) on the spawning migration to natal tributaries. Average abundance of Cultus sockeye salmon declined from over 20,000 spawners from 1930 to 1960, to less than 10,000 from 1960 to 1990, and less than 4,000 since 1990. The Committee on the Status of Endangered Wildlife in Canada designated the population as *Endangered* in 2003 (COSEWIC 2003), and Fisheries and Oceans Canada established a recovery team in 2004 (CSRT 2009). Factors responsible for low survival have been identified in both the freshwater and marine environments occupied by Cultus sockeye salmon (CSRT 2009). Some of these factors are specific to the Cultus population whereas others also impact other Fraser River late run sockeye salmon populations (Cooke et al. 2006; Miller et al. 2011).

Even prior to the most recent declines in abundance, the Cultus sockeye population was characterized by unusual allele frequencies and low levels of diversity at allozyme and microsatellite loci relative to other Fraser River sockeye populations (Withler et al. 2000). A primary objective of the conservation strategy developed for the Cultus sockeye population was to "Ensure the genetic integrity of the population by exceeding a 4-year arithmetic mean of 1,000 successful adult spawners with no fewer than 500 successful adult spawners on any one cycle" (CSRT 2009). Conservation measures have been undertaken to improve both freshwater and marine survival, including annual reduction of Eurasian watermilfoil along the lakeshore, predator (northern pikeminnow) removal from the lake limnetic zone and harvest restrictions in the mixed-stock Fraser River sockeye salmon fisheries (Bradford et al. 2011). In addition, a two-generation (8 year) captive breeding program was initiated to protect a random sample of fish from the extremely low and unpredictable smolt-to-adult survival that threatened the population's persistence, and to promote the production of large numbers of juvenile fish for release into Cultus Lake or its outlet stream, Sweltzer Creek.

In this study, we use genetic methods to assign individuals that have returned from hatchery releases of Cultus Lake sockeye salmon to their parents and grandparents and estimate their potential contribution to the wild population. We establish a pedigree for the first generation of large-scale hatchery production that will be used in this initial

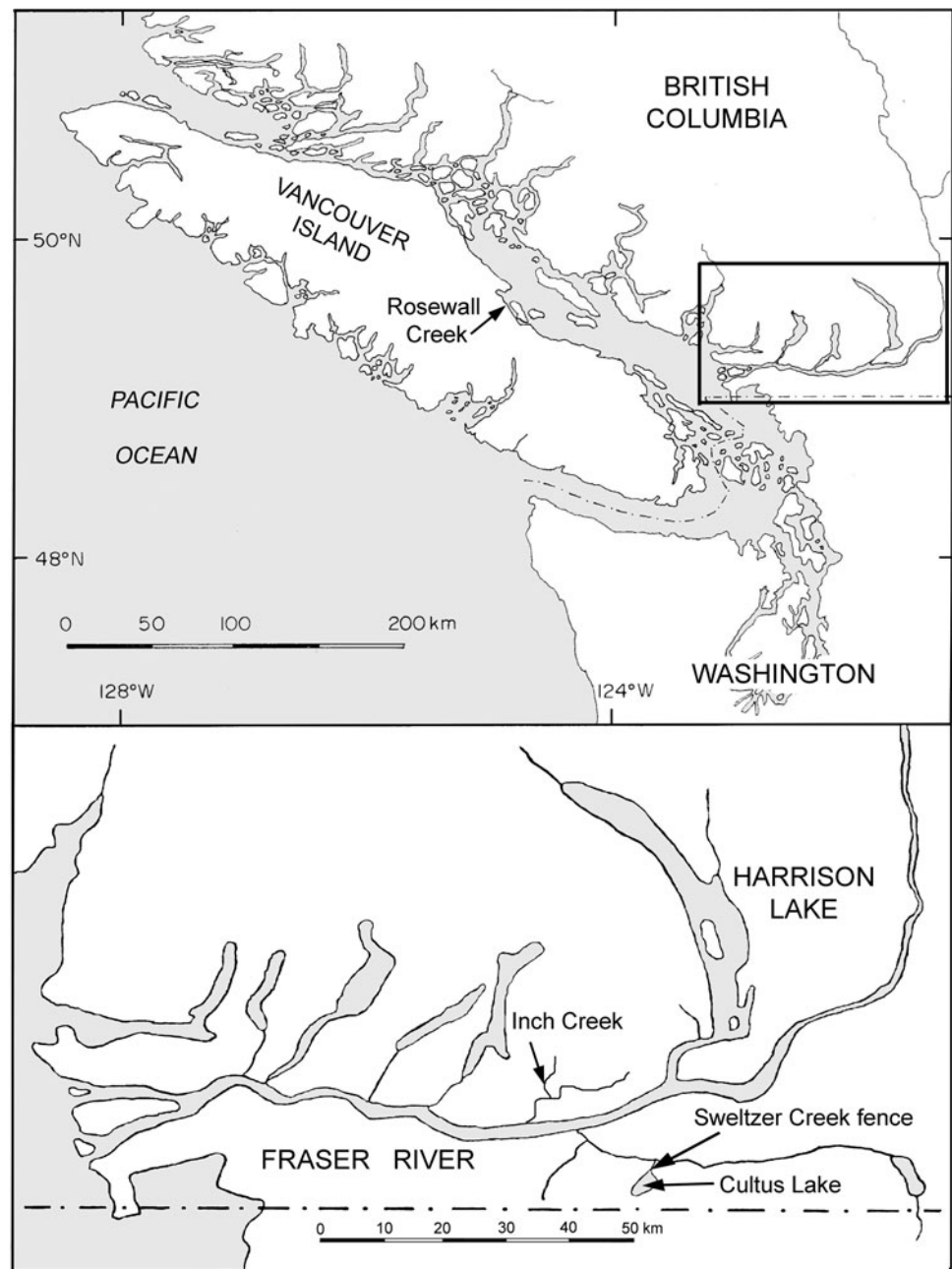
and ongoing assessment of the impacts of hatchery production on effective population size and coancestry among Cultus sockeye salmon that spawn in the wild. Our goal is to determine whether the hatchery program makes a beneficial contribution to wild population persistence, a contribution requiring the successful reproduction of hatchery-produced fish in the natural environment of Cultus Lake.

The Cultus hatchery program, established in 2000, was founded with wild-born indigenous sockeye salmon. Spawning of broodstock has been conducted at the Cultus Lake laboratory with egg incubation and juvenile rearing undertaken at the nearby Inch Creek Hatchery (ICH) in the lower Fraser River drainage (Fig. 1). Most of the juveniles from ICH were released into Cultus Lake the year following spawning (hatchery supplementation), but a subset of juveniles was transferred each year to Rosewall Creek Hatchery (RCH) on Vancouver Island for rearing until maturity and spawning (captive breeding). In addition, wild Cultus sockeye salmon smolts emigrating from Cultus Lake were captured in 2002 and 2003 and used to augment the fish transferred to RCH for the captive breeding. Once captive breeding at RCH started, juvenile progeny of captive breeders were transferred back to ICH, combined with the juveniles produced from supplementation and released to Cultus Lake or Sweltzer Creek as juveniles (Fig. 2).

Large numbers of captive-bred progeny were first produced at RCH in 2004, a year in which fewer than 100 wild sockeye salmon returned to Cultus Lake itself, the lowest return on record (since 1925). The release of large numbers of captive-bred juveniles from a year in which few wild sockeye had entered Cultus Lake to spawn raised concern about the level of genetic diversity that would be present in the 2008 Cultus sockeye salmon return, expected to comprise predominantly hatchery-produced age four fish. In the current study we examined the level and sources of genetic variation in the hatchery-produced sockeye salmon that returned from the 2004 captive spawning to Cultus Lake in 2007/2008 to determine the total genetic contribution to the population from a single year of captive breeding. We performed the same analyses on all Cultus Lake sockeye salmon that returned to spawn in 2008 to determine the potential genetic impact of captive breeding on a single year of natural spawning when few wild fish returned.

A two-generation pedigree was established for the hatchery program with data from 14 polymorphic microsatellite loci. The microsatellite genotypes were used to determine parent-progeny relationships between the 2004 hatchery broodstock fish (generation G_1) and their adult progeny that returned in 2007 and 2008 (G_2). In addition, relationships among G_1 genotypes were used to reconstruct the prior generation (G_0) that produced them. The pedigree enabled identification of dominant wild-born and hatchery-born ancestors that contributed to G_2 . This in turn allowed

Fig. 1 Location of Cultus Lake and the Inch Creek (ICH) and Rosewall Creek (RCH) hatchery facilities in southern British Columbia, Canada



estimation of the genetically effective number of mature sockeye salmon that returned to Cultus Lake in 2007 and 2008 from the 2004 hatchery program and of the total effective number of breeders (N_b) in the 2008 return of 521 sockeye salmon to Cultus Lake.

Materials and methods

Sampling of Cultus Lake sockeye salmon in the wild

Cultus sockeye salmon spawn on gravel beds associated with the beaches and bays of Cultus Lake. Juvenile fish

(smolts) emigrate through the lake outlet (Sweltzer Creek) and the lower Fraser River drainage to its the estuary in Georgia Strait (Fig. 1). The marine migratory route involves northward movement in coastal BC waters, offshore feeding in the Gulf of Alaska and a return trip undertaken as maturation approaches at age three, four or, less frequently, five (CSRT 2009). Thus, generations overlap in the wild, with spawning fish in each year generally derived from at least 2 years of prior spawning.

Wild (i.e. naturally-spawned) sockeye salmon returning to Cultus Lake in 1992 and from 1999 to 2008 were sampled for genetic analysis at the Sweltzer Creek fence, after transfer from the fence into the broodstock program,

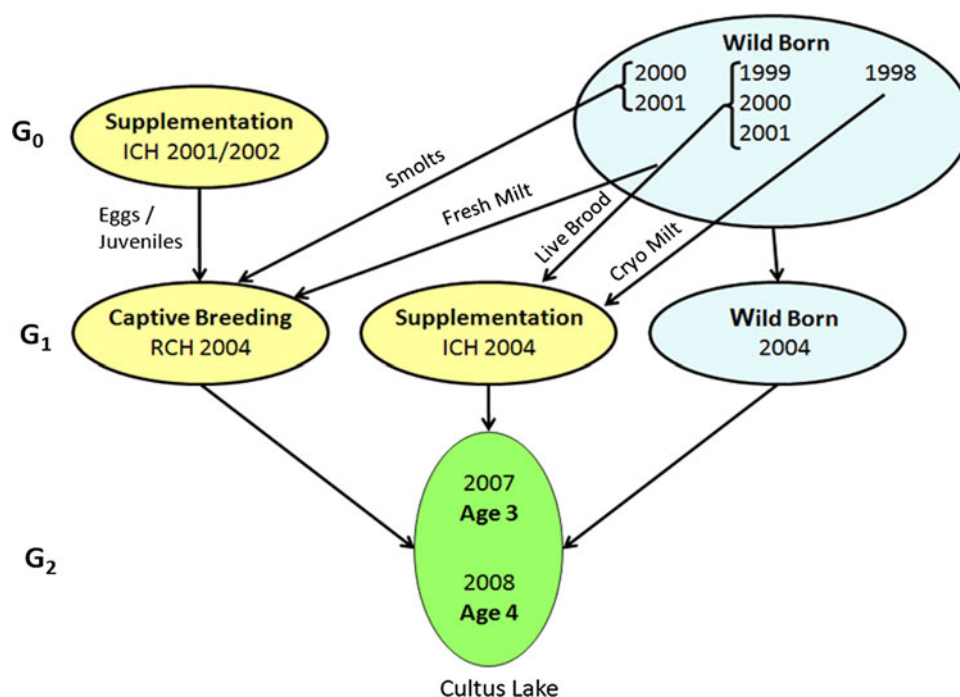


Fig. 2 Origins of three pedigreed generations in a breeding program for Cultus Lake sockeye salmon. G₀ fish (grandparents) provided G₁ progeny that were both wild-born (between 1998 and 2001) and hatchery-born from supplementation (crosses made at Cultus Lake lab in 2001 and 2002). At maturity in 2004, G₁ fish (parents) were spawned at both Cultus Lake laboratory (supplementation ICH) and RCH (captive breeding). Both hatchery-born and wild-born G₁ were

spawned at RCH, whereas only wild-born G₁ were spawned at the Cultus Lake lab. G₂ fish (progeny) were offspring of G₁; they were released into Cultus Lake as juveniles and sampled as returning adult fish in 2007 (at age three) and 2008 (at age four). Of 521 sockeye salmon that returned to Cultus Lake in 2008, approximately 40 were wild-born fish and the remainder were hatchery-produced

or post-spawning in Cultus Lake. Except for 1992, when liver tissue was collected, samples were small punches of operculum tissue, labelled and stored individually in 95% undenatured alcohol. Except for some hatchery progeny produced in 2004, all hatchery-produced Cultus Lake sockeye salmon have been marked with an adipose fin clip. From 2005 on, marked hatchery-produced fish returned to Cultus Lake and were sampled in proportion to their abundance. In 2007 and 2008, tissue samples were collected from 35% (281 of 800) and 50% (260 of 521) of the adult sockeye salmon returns to Cultus Lake.

Hatchery spawning and sampling

Reared in captivity at RCH, the majority of Cultus Lake sockeye salmon (both male and female) mature at age three, with small proportions maturing at ages two and four. Thus, the transfer of juveniles to RCH for rearing until maturity and re-introduction of their progeny to Cultus Lake has increased the overlap of generations in Cultus Lake wild spawning. The hatchery- and wild-born fish spawned at RCH and Cultus Lake in 2004 comprised G₁ of this study and were sampled for genetic analysis. At Cultus Lake, only 25 live wild-born fish (10 females and 15 males) were spawned, but cryopreserved

milt from 36 additional wild-born males that had returned to Cultus Lake in 2002 was also used in egg fertilization. At RCH, 495 spawners were hatchery-born. They comprised 486 age three progeny of 20 G₀ fish spawned at Cultus Lake in 2001 and nine age two progeny from the 2002 Cultus Lake crosses. Seventeen of the 20 fish spawned in 2001 were sampled for genetic analysis.

Wild-born G₁ fish spawned at RCH in 2004 were 236 fish that matured from the wild smolt collections made in 2002 and 2003. Additionally, milt from 12 of the 15 live wild-born brood fish used at Cultus Lake in 2004 was also transferred to and used in crosses made at RCH. Crosses at RCH were made both within and between the hatchery- and wild-born parents, with parents of both sexes generally mated with two or three individuals. Only two-thirds of the hatchery-produced fish were marked, with all unmarked fish progeny of hatchery-born parents spawned at RCH between Nov. 18 and 23. At Cultus Lake, eggs of the 10 wild female broodstock were fertilized in semi-factorial fashion with milt from the 15 live males and cryopreserved milt from the 36 males from 2002. Virtually all (>95%) of the live and cryopreserved G₁ parental spawners used at both RCH and Cultus Lake laboratory in 2004 were sampled for genetic analysis.

Microsatellite analysis

Genetic analysis of archived tissue samples collected from fish sampled in the wild was begun in 2006. Analysis of archived samples from hatchery spawners was not undertaken until unusual allele frequencies were observed in the hatchery-produced fish that returned to Cultus Lake in 2007 (at age three) from captive breeding in 2004. All genetic analyses were conducted on a dataset of 14 polymorphic unlinked microsatellite loci, all of which display Mendelian single locus inheritance in the Cultus sockeye salmon population as described in Beacham et al. (2005). The loci were Ots2, Ots3 (Banks et al. 1999), Ots100, Ots103, Ots107, Ots108 (Beacham et al. 1998; Nelson and Beacham 1999), Oki1a, Oki1b, Oki6, Oki10, Oki16, Oki29 (Smith et al. 1998; Nelson et al. 2003), One8 (Scribner et al. 1996), and Omy77 (Morris et al. 1996). Samples scored for less than 10 loci were excluded. Allelic richness (AR) standardized to a sample size of 24 fish, gene diversity (H_E) and pairwise F_{ST} values were calculated for samples using FSTAT (Goudet 1995, 2001). The ‘two parent’ exclusion power of the microsatellite loci for Cultus sockeye salmon was determined (Jamieson and St. Taylor 1997).

Parentage assignment and pedigree reconstruction

The two generation pedigree established for the progeny (G_2 generation) that returned in 2007 and 2008 from the 2004 spawners (G_1 generation) includes grandparents (G_0 generation) that spawned in the wild between 1998 and 2001 as well as those that were hatchery-spawned at Cultus Lake in 2001 and 2002. No tissue samples were available for G_0 grandparents that spawned in the wild. Instead, the identification of full- and half-sibling families among the wild-born G_1 spawners, conducted by maximum likelihood (ML) analysis in the program COLONY (Wang 2004), enabled their assignment to G_0 parental fish. The fish collected as smolts in 2002 and 2003 were all assumed to have been emigrating at age one; therefore they arose from wild spawning that occurred in 2000 and 2001, respectively. The G_1 broodstock fish spawned at Cultus were assigned to wild G_0 parents that spawned between 1999 and 2001 based on their ages (3–5) as determined by otolith readings. Thus, wild fish spawned at Cultus in 2004 shared some G_0 parentage from 2000 and 2001 with the smolts, collected from the wild, that matured and were spawned at RCH in 2004. Genotypes for 739 and 254 wild-born fish from the 2000 and 2001 spawning years were analyzed, respectively. Of these, the wild-born fish that were spawned at RCH and the Cultus Lake lab in 2004 were 223-fish and 30-fish subsets from 2000 and 2001, respectively.

The ML analysis conducted with COLONY (Wang 2004) consisted of the simultaneous examination of all individuals from each spawning year (a group-likelihood approach) in order to partition them into families of variable size based on Mendelian inheritance of alleles at unlinked loci. The program was run assuming parental polygamy and with both types of genotyping error set at 0.01 for all loci. Error types are class I (allelic dropout) and class II (all other ‘errors’ including miscalling of alleles, PCR artifacts and mutation). The assigned error rate was considered an upper bound for plausible values given that error rates in repeated genotyping of samples from this species in our laboratory are almost an order of magnitude lower. Re-genotyping does not reveal all sources of Mendelian ‘error’ (e.g. mutation, some PCR artifacts). For loci as polymorphic as the Cultus Lake sockeye salmon microsatellites of this study, the probability of including an unrelated individual in a fullsib family of five or greater was likely less than 0.01 (Almudevar and Field 1999). Since 70% of sampled wild-born fish from 2000 and 2001 were included in full sibships of five or more fish (unpublished data), the majority of the wild-born G_1 fish were likely identified correctly to G_0 family.

Tissue samples were available for most of the hatchery-born grandparents spawned in 2001, but not 2002. Thus, all 495 hatchery-born spawners of 2004 except the nine born in 2002 were assigned to their parents by genotypic exclusion. Assignment of marked G_2 fish to the large number of G_1 parents, also by exclusion, was facilitated by the use of CERVUS (Kalinowski et al. 2007) and of hatchery records. Unmarked fish that returned from the 2004 spawning were also assigned to hatchery parents by exclusion if possible; only those that could not be assigned hatchery parents were classified as wild. A summarized G_0 – G_2 pedigree was drawn with the assistance of the PedigraphTM program (Garbe and Da 2008).

Analysis of gene origin and effective population size

Age three and four fish that returned to Cultus Lake in 2007 and 2008 were the generation of interest (G_2) in this study. Assessment of this generation enabled calculation of the effective potential genetic contribution from the first year of large-scale captive breeding to the Cultus sockeye population, with the contribution spread over 2 years (there was a negligible return of age five fish in 2009). An analysis of probable gene origins (Boichard et al. 1997) and estimation of effective population size (N_e) in the Cultus sockeye G_1 and G_2 generations was undertaken through pedigree analysis conducted with the program ENDOG 4.5 (Gutiérrez and Goyache 2005).

Average relatedness (A_R), the probability that an allele chosen randomly from the whole population in the

pedigree belongs to a given individual, was calculated for all pedigreed individuals. Ancestors in a pedigree are individuals who have made non-redundant genetic contributions; the numbers of total, effective and ‘accountable for 50% of genetic variation’ ancestors were calculated for both populations of interest (Boichard et al. 1997). Founders are ancestors whose parents are unknown and founder genome equivalents (FGE) are the number of founders required to produce the observed level of genetic diversity if all had contributed equally with no loss of alleles (Ballou and Lacy 1995). In ENDOG, FGE values are calculated as the inverse of twice the average coancestry of the individuals in the population (Caballero and Toro 2000).

Inbreeding effective population size, N_{ei} , of G_2 was calculated in ENDOG using the increase in inbreeding (ΔF) in each generation in the relationship $N_{ei} = 1/2\Delta F$. Note that because there were no known relationships among fish of G_0 , no ΔF took place in G_1 by definition; hence the estimated values of N_{ei} account only for the ΔF that occurred in G_2 . Variance effective population size (N_{ev}) was calculated for G_1 and G_2 using the variance in progeny production according to: $N_{ev} = (\sum N_{ij})^2 / \sum N_{ij}^2$ where N_{ij} are the numbers of progeny produced by male and female parents. The N_{ev} value over two generations was calculated as the harmonic mean of G_1 and G_2 values. Importantly, N_e values calculated were not N_e values for the Cultus Lake sockeye salmon population as a whole because, for a Pacific salmonid population, N_e is the sum of the effective number of breeders (N_b) in each year of the average generation time (Waples 2004).

The G_2 returned to Cultus Lake over 2 years and constituted the majority of the fish that returned in 2008. The impact of the hatchery program on a single yearclass of the wild population was evaluated by extending the analysis of gene origin and effective population size to all adult sockeye salmon returns to Cultus Lake in 2008 (thus including 3-year old fish derived from 2005 spawning).

Calculation of contributions from 2005 wild and hatchery spawners were conducted in the same manner as described for 2004.

Maximum likelihood analysis of parental contributions

The restricted parentage, and resulting unusual allele frequencies, of the hatchery-born G_1 at RCH in 2004 enabled an independent estimation of their genetic contribution to the returning G_2 fish sampled in 2007 and 2008. Admixture of genes contributed by G_1 hatchery- and wild-born parents to G_2 was estimated from the microsatellite data using the likelihood estimation of admixture (LEADMIX) program (Wang 2003). This analysis is based on a model in which two or more source populations derived from a common ancestral population become admixed (hybridized) after secondary contact is re-established. Changes in allele frequencies through drift occur both in the source populations after their divergence from the ancestral population and in the admixed population derived from the source populations. The proportions of genes contributed by the source populations (and their 95% confidence intervals) are estimated based on an assigned minimum level of drift (0.00001 herein), with mutation ignored.

Results

G_2 fish, resulting from hatchery spawning in 2004, returned to Cultus Lake in 2007 and 2008 (Table 1). They were the first hatchery fish to make a significant contribution to total adult returns. Small numbers of hatchery fish had returned from supplementation crosses in 2005 and 2006 (Table 2), but returns in those years were composed of 99 and 80% wild fish, respectively. In contrast, 40% of the 2007 return and 93% of the 2008 return was hatchery-produced. In 2008, 90% of the hatchery fish were from the 2004

Table 1 Age, mark status and birthplace (hatchery or wild) of sockeye salmon in the 2007 and 2008 adult returns to Cultus Lake

Return year	<i>n</i>	Marked	Unmarked	Hatchery	Wild
2007					
Age three	62	56	6	62 (G_2)	0
Age four	219	50	169	50	169
Total	281			112	169
2008					
Age three	26	23	3	23	3
Age four	234	198	36	218 (G_2)	16
Total	260			241	19

Sample size is given as *n*. Age three and four fish belonging to G_2 are indicated. In 2004 only, approximately one-third of hatchery fish were not marked before release. Unmarked hatchery fish that returned at age three in 2007 or age four in 2008 were identified by assignment to hatchery parents and included with marked fish in the hatchery-produced G_2 numbers

Table 2 Standardized (to sample size 24) allelic richness (AR), genic diversity (H_E —expected heterozygosity) and observed heterozygosity (H_O) over 14 microsatellite loci in wild-born and hatchery-produced Cultus Lake sockeye salmon of sample size n

Year	n	AR	H_E	H_O
Wild				
1992	45	6.3	0.60	0.58
1999	76	6.1	0.59	0.61
2001	58	5.9	0.59	0.61
2002	172	6.2	0.59	0.58
2004	28	6.3	0.60	0.61
2005	151	6.1	0.60	0.59
2006	504	6.2	0.60	0.59
2007	169	6.1	0.59	0.58
Hatchery returns				
2006 (4-year-old)	72	6.2	0.60	0.59
2007 (4-year-old)	50	5.8	0.60	0.61
2007 (3-year-old G_2)	62	5.6	0.57	0.58
2008 (4-year-old G_2)	218	6.0	0.58	0.59
Hatchery spawners				
2004 hatchery-born (G_1)	495	4.5	0.52	0.55
2004 wild-born (G_1)	297	6.2	0.60	0.60

Fish belonging to G_1 and G_2 are indicated

hatchery spawning, only a few were age three fish from hatchery spawning in 2005 (Table 1).

The total number of hatchery-produced G_2 fish sampled in this study was 280, 62 at age three and 218 at age four (Table 1). In 2007, all 56 marked and all six unmarked age three fish that were sampled in the adult return were assigned by exclusion to 2004 spawners at RCH or Cultus Lake. In 2008, the total adult return to Cultus Lake was 521 fish, of which half (260 fish) was sampled for genetic analysis. Of the 260 sampled fish, 234 and 26 fish were age four and age three, respectively (Table 1). Among the age four fish, all 198 marked and 20 of 36 unmarked fish were assigned to 2004 hatchery production. The unmarked fish assigned to G_1 hatchery parents were members of the unmarked hatchery production from the 2004 spawning year. No hatchery parents were identified for the remaining 16 unmarked age four fish sampled in 2008; these were therefore classified as wild progeny of the 53 adult sockeye salmon that entered Cultus Lake to spawn in 2004.

Of the 26 age three fish that were sampled in the 2008 return, only three were wild (unmarked). Thus, only 19 (three age three and 16 age four) wild fish were identified in the half of the 2008 return that was sampled and it is likely that the number of wild fish in the total return of 521 fish was approximately 40. Therefore, wild fish constituted less than 10% of the Cultus sockeye return in 2008.

Allelic diversity

Genic diversity and allelic richness at the 14 microsatellite loci did not change in wild Cultus sockeye salmon sampled

between 1992 and 2007 (Table 2). Hatchery-produced fish from 2002 and 2003 (primarily supplemental releases from ICH) had diversity values similar to those of wild samples (Table 2). However, G_2 hatchery-produced age three fish sampled in 2007 (the captive-bred fish of this study) had lower levels of both genic diversity (H_E) and allelic richness (AR). Allele frequencies of these fish were also unusual compared to other samples of naturally- and hatchery-spawned Cultus sockeye salmon. Whereas pairwise F_{ST} values between samples of wild fish and prior hatchery returns were less than 0.01 and seldom significant, pairwise F_{ST} values involving the captive-bred hatchery-produced fish from 2004 with other groups were all significant ($P < 0.05$) and ranged up to 0.02.

The reduced genetic variation observed in the hatchery-produced age three fish of 2007 prompted the genetic analysis of the 2004 G_1 hatchery spawners that produced them. This analysis revealed that the hatchery-born spawners that dominated the RCH broodstock had also been characterized by low levels of diversity, skewed allele frequencies and a significant excess of observed heterozygotes compared with those expected (Table 2). These fish, progeny of crosses made at Cultus Lake in 2001, were primarily members of two large halfsib families (Families A and B), each resulting from the mating of a single female with two different males (Table 3). The remainder were progeny of a factorial cross among five females (P–T) and five males (V–Z) (Table 3). However, values of both AR and H_E were higher in the age four G_2 fish that returned in 2008 than in the age three G_2 (Table 2).

Table 3 Composition of the grandparental (G_0) and parental (G_1) generations at Cultus Lake laboratory and Rosewall Creek Hatchery (RCH) that produced the age three and four G_2 sockeye salmon that returned to Cultus Lake in 2007 and 2008

G ₀ date	G ₀ number		G ₁ description	G ₁ number			
				RCH 2004		Cultus 2004	
	Male	Female		Male	Female	Male	Female
Wild-born							
1998	2	2	Cultus cryo milt	–	–	36	0
1999	5	5	Cultus age 5	–	–	6	2
2000	80	78	RCH smolts; Cultus age 4	10	203	6	4
2001	26	29	RCH smolts; Cultus age 3	17	6	3	4
Total	227			297			
Hatchery-born							
2001	A1	A	RCH family A1	54	77	–	–
2001	A2	A	RCH family A2	64	85	–	–
2001	B1	B	RCH family B1	33	38	–	–
2001	B2	B	RCH family B2	30	38	–	–
2001	PQRST	VWXYZ	RCH 5 × 5 families	31	36	–	–
2002	3	3	RCH age two males	9	0	–	–
Total	22			495			

These fish are also shown in the pedigree summary (Fig. 3)

Reconstruction of G_0 generation

Assignment of G_1 fish (2004 hatchery spawners) to parents reconstituted the grandparental (G_0) generation fish that spawned in the wild or were hatchery-spawned at Cultus Lake (Table 3; Fig. 3). The 297 wild-born G_1 spawners at RCH and Cultus in 2004 originated from 227 G_0 parents that had spawned between 1998 and 2001, inclusive. The majority of them (207 females and 16 males) were age four fish that were among 739 fish arising from wild spawning in 2000 that were genotyped and assigned to 155 full-sib families. The 223 G_1 spawners wild-born in 2000 included broodstock used both at Cultus and RCH and were assigned to 158 G_0 parents (Table 3; Fig. 3); each of these parents had spawned in the wild with as many as three contributing mates and individually produced up to 17 progeny in G_1 .

Similarly, the 30 age three wild-born G_1 spawners used at RCH and Cultus were assigned to 55 G_0 parents from 2001 and the eight age five Cultus spawners were assigned to ten G_0 parents from 1999 (Table 3; Fig. 3). The family structure of 36 males that contributed milt that was cryo-preserved in 2002 and used at Cultus in 2004 was unknown, but only four of them produced progeny that returned in 2007/2008. These four males were arbitrarily assigned to two single pair families (four parents) from 1998 (Table 3; Fig. 3).

Assignment of hatchery-born G_1 spawners used at RCH in 2004 to G_0 crosses made with 20 parents in 2001 at Cultus Lake was straightforward due to the large number of

G_1 progeny involved. Four of the 20 G_0 parents produced no G_1 progeny that were spawned in 2004; these four parents have not been shown in the G_0 (Table 3; Fig. 3). The remaining 16 G_0 parents hatchery-born in 2001 are each indicated by a letter or letter-number combination (Table 3; Fig. 3). The vast majority of hatchery-born G_1 spawners were members of Families A and B, created in 2001 with six parents (Tables 3, 4; Fig. 3).

The nine G_0 males from 2002 Cultus crosses were not sampled and therefore were arbitrarily assigned to three hatchery-produced families (three males to each pair of parents) (Table 3; Fig. 3). Thus, the 495 hatchery-born spawners of G_1 arose from a total of 22 G_0 parents, 16 spawned in 2001 and six in 2002 at Cultus Lake. The hatchery-born spawners at RCH, especially Families A and B, provided most of the gametes in G_1 hatchery production of 2004 (Table 4). Most G_1 parents spawned at RCH and Cultus in 2004 contributed to progeny production in G_1 although many did not contribute to subsequent G_2 adult returns in 2007 and 2008.

Assignment of G_2 individuals to G_1 parents and ML analysis of G_2 allele frequencies

Based on allele frequencies of wild Cultus Lake sockeye salmon from 1992 to 2005 combined, the two-parent exclusion power of individual microsatellite loci ranged from 0.13 to 0.97, with a combined (over the 14 loci) value of >0.99. However, because of the large number of G_1

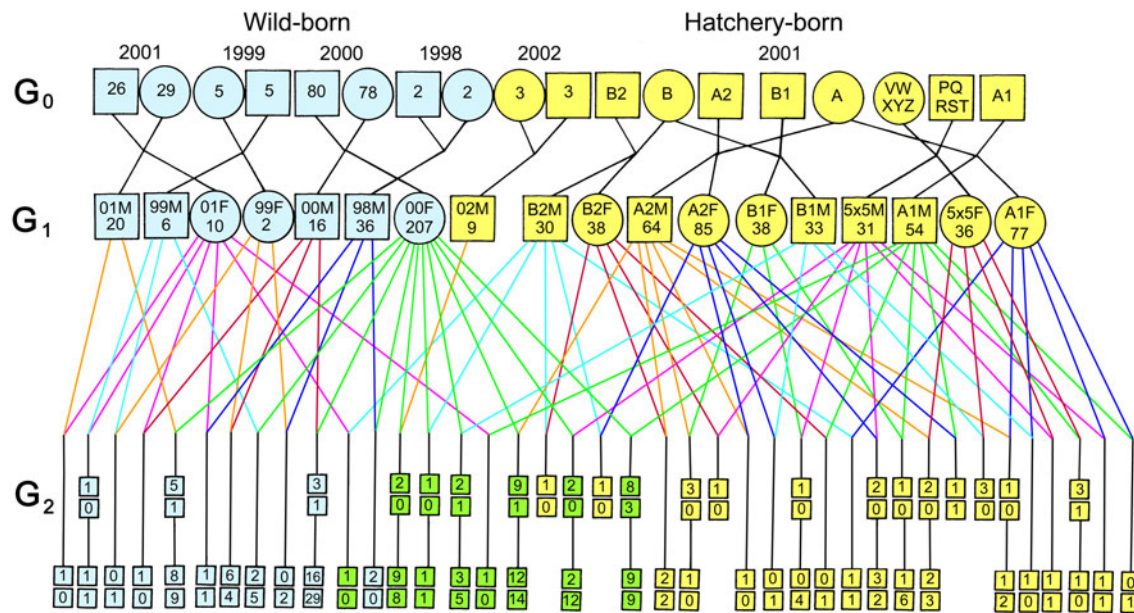


Fig. 3 Pedigree summary for three generations (G_0 – G_2) of Cultus Lake sockeye salmon. In the *bottom rows* of the pedigree are shown G_2 sockeye salmon that returned to Cultus Lake at age three in 2007 (*top row of paired boxes*) and age four in 2008 (*bottom row of paired boxes*). In each pair of boxes, the numbers of G_2 males (*top box*) and females (*bottom box*) are shown. G_2 fish were progeny of G_1 hatchery crosses made in 2004 with wild-born and hatchery-born sockeye salmon at RCH and Cultus Lake (*middle row of pedigree*) and grandchildren of wild-born G_0 fish that spawned in the wild or were spawned in the hatchery at Cultus Lake lab (*top row of pedigree*). Fish in G_2 were progeny of two wild-born parents (*blue shaded boxes*), of

two hatchery-born parents (*yellow shaded boxes*) or of one wild-born and one hatchery-born parent (*green shaded boxes*). In G_0 and G_1 , female groups are depicted in *shaded circles* and male groups in *shaded squares*. Groups of wild-born G_1 fish and their G_0 parental groups are indicated by egg fertilization (birth) year and shaded in *blue*, with sample sizes indicated within *circles* and *squares*. Groups of hatchery-born G_1 fish and their individual G_0 parents, indicated by *letters* or *letter–number combinations*, are shaded in *yellow*. Male and female sample sizes for hatchery-born G_1 groups are given within *squares* and *circles*, respectively

Table 4 Numbers of Cultus Lake sockeye salmon G_1 and their percentage gametic contributions to fertilized eggs produced at RCH and Cultus Lake in 2004 and to their age three and four adult progeny G_2 that returned to Cultus Lake in 2007/2008

G_1 Broodstock source	G_1 Parents	% Gametes in 2004	% Genes in 2007 age 3 G_2	% Genes in 2008 age 4 G_2
Hatchery-born				
RCH (Fam A1, A2)	280	35	32	20
RCH (Fam B1, B2)	139	22	13	7
RCH (5 × 5 crosses)	67	9	12	8
Hatchery (2002 males)	9	3	2	4
Hatchery-born				
Assign	495	70	59	39
ML			59 (49–68)	33 (29–40)
Wild-born				
RCH mature smolts	236	24	38	45
RCH (milt—Cultus brood)	(12) ^a	3	1	6
Cultus live broodstock	25	3	2	9
Cultus (2002 frozen milt)	36	1	0	1
Wild-born				
Assign	297	30	41	61
ML			41 (32–51)	67 (60–71)

The contributions from hatchery-born and wild-born G_1 parents were estimated by both parentage assignment and ML analysis of allele frequencies. Confidence intervals (95%) for the ML estimates are given in parentheses

^a Milt from 12 of 15 males spawned at Cultus Lake Laboratory in 2004 was also used at RCH (in parentheses)

crosses made in 2004 with fish belonging to Families A and B (Fig. 3), more than one two-parent matches were identified for some G_2 fish from these crosses. In all cases, use of the cross records enabled identification of the two most likely parents.

Assignment of 62 age three fish that were sampled in 2007 to their wild- and hatchery-born G_1 parents indicated that they returned approximately in proportion to their production in 2004 (Table 4). The six G_0 grandparents from Families A and B, who produced over half of all G_1 spawners (Tables 3, 4), contributed 45% of the genes to the 2007 G_2 fish. The 16 successful grandparents spawned at Cultus in 2001 contributed 59% of the genes to them (Table 4). Thus, 47 of the 62 G_2 fish had one or both G_1 parents among the 419 spawners in Families A and/or B. Only one age three G_2 fish was assigned to wild broodstock spawned at Cultus in 2004; the remainder all came from RCH crosses, although one was sired by one of the wild Cultus male broodstock fish whose milt was also used to fertilize eggs at RCH (Table 4). ML estimation of the proportions of genes contributed by the hatchery-born parents spawned at RCH in 2004 and the wild-born fish spawned at RCH and Cultus Lake combined to the fish that returned in 2007 were in good agreement with the proportions estimated based on parentage assignment (Table 4).

The contribution of G_1 hatchery-born parents, particularly those of Families A and B, to the age four G_2 sampled in 2008 was less than to the age three fish of the previous year (Table 4; Fig. 3). Based on assignment, the hatchery-born spawners cultured at RCH accounted for only 39% of genes in the age four fish; the contributions of wild-born spawners at both Cultus Lake and RCH increased correspondingly (Table 4). In particular, the crosses made with live broodstock at Cultus accounted for 9% of returning 2008 adult progeny, although they were responsible for less than 5% fish released from G_1 crosses. ML estimation of

parental contributions indicated a slightly lower (33%) contribution from hatchery-born parents to the age four G_2 (Table 4). Thus, while the age three G_2 were primarily progeny of the hatchery-born G_1 that was derived from 22 G_0 fish, the age four G_2 were primarily progeny of the wild-born G_1 derived from 227 G_0 grandparents (Table 3; Fig. 3).

Analysis of inbreeding and gene origin

Average relatedness based on all pairwise estimates between fish for G_1 was 4.2% (Table 5). Most of the increase in relatedness in this generation was attributable to the large numbers of full- and half-sib progeny among the hatchery-born G_1 . A_R was lower in G_2 than G_1 , especially among the age four fish that returned in 2008 (Table 4). Nine inbred G_2 fish returned in 2007 and 2008, all resulting from mating of full- and half-sib G_1 parents belonging to Families A and B (Tables 4, 5). Six of the inbred G_2 fish had one parent each in the two half-sibling subsets of Family A (A1 and A2). Three of them resulted from crosses in which the parents were full-sibs; one each from Families A1 and A2 and one from Family B1. Thus, the mean inbreeding in G_2 was low (approximately 1% among the age three and <0.5% among age four fish). Most fish were not inbred, and those that were had ΔF values of 12.5 or 25% (Table 5).

A total of 142 ancestors was identified for the 280 G_2 fish sampled in 2007 and 2008 (Table 5). However, the numbers of ‘effective’ ancestors and ancestors that accounted for 50% of the genetic material in G_2 were much lower and all numbers were lower in G_1 (Table 5). Dominant ancestors for G_1 were, as expected from the pedigree, the six G_0 parents that produced Families A and B reared at RCH (Tables 3, 4). Together they accounted for over half of the genetic content of G_1 and also made a major contribution (31%) to G_2 . Eleven of the 20 ancestors that

Table 5 Analysis of gene origin, inbreeding and effective population size in Cultus Lake sockeye salmon (both for the age three and four fish that comprised the adult G_2 from G_1 spawners bred in 2004 and for the total breeding population that returned to Cultus Lake in 2008)

Generation	<i>n</i>	Ancestors			Related %	Mean % ΔF	% Inbred fish	<i>n</i> (and % ΔF) of inbred fish	N_{ei}	N_{ev}	Mean N_{ev}	FGE
		Total	Eff	50%								
G_1	792	202	15	5	4.2	0	0	0	–	32	–	15
G_2												
3-Year-old	62	44	18	7	3.8	1.01	6.45	3 (12.5) 1 (25.0)	50	67	–	15
4-Year-old	218	131	42	22	2.4	0.40	2.29	3 (12.5) 2 (25.0)	125	157	–	36
Total G_2	280	142	36	20	2.7	0.54	3.21	6 (12.5) 3 (25.0)	93	183	54	32
Total 2008	242–260	173	58	33	1.4	0.34	1.92	3 (12.5) 2 (25.0)	138	193	72	48

For groups of G_1 and G_2 fish, the numbers sampled (*n*), numbers of ancestors (total, effective and number required to account for 50% of genes present) and average level of relatedness are given. For G_2 , mean increase in inbreeding, percentage of inbred individuals and numbers and levels of inbreeding of inbred individuals are shown. Variance (N_{ev}) and inbreeding (N_{ei}) effective population size estimates and the harmonic mean of N_{ev} for G_1 and G_2 are given. Estimates of effective numbers of FGE for G_1 and G_2 are shown

contributed 50% of the genetic material in G_2 (Table 5) were G_0 grandparents spawned at Cultus in 2001 (Table 3). Three of the remaining top contributing ancestors to G_2 , however, were among the 25 wild broodstock collected and spawned at Cultus in 2004. The remaining six dominant ancestors of G_2 were G_1 fish from the wild smolt collections that were spawned at RCH in 2004, or G_0 parents thereof.

Effective population size

For all of the pedigreed G_2 fish in the returns from the 2004 hatchery production, estimated N_{eI} was 93 (Table 5). N_{eI} was larger for the more diverse age four returns of 2008 (125) than for the more inbred age three returns of 2007 (50). The N_{eI} estimates were larger than those based on variance in progeny production (Table 5) because of the shallowness of the available pedigree (which precluded an estimate of ΔF in G_1) and the fact that fewer inbred G_2 progeny returned than expected based on parental gametic contributions in 2004, especially in 2008.

The G_1 N_{eV} value of 35 revealed the effect of the pre-dominance of fish belonging to Families A and B in G_1 (Table 3). This number is the ‘realized’ N_{eV} , because it is based only on G_1 fish that reproduced successfully (i.e. parented a fish that returned in G_2). In contrast, the N_{eV} of 183 estimated for G_2 (Table 5) has not yet been realized, in that it is based on fish that survived to reproductive age in 2007 and 2008, but have themselves not yet proven reproductively successful. The harmonic mean of N_{eV} for G_1 and G_2 was 54.

Estimates of N_e calculated for all fish in the breeding population of 2008 (N_b) were higher than those for the age four hatchery-produced fish due to the presence of 23 pedigreed hatchery-produced 3-year-old fish from spawning in 2005 (Table 4). In addition, the estimates of N_{eV} , but not N_{eI} , for 2008 were increased by the presence of low numbers of wild age three and four fish. The wild fish, for which a two-generation pedigree was not available, were by definition founders and not inbred. Inclusion of the additional fish increased the N_{eV} estimates for both G_1 (44) and G_2 (193), providing a harmonic mean value of 72 for the 2008 N_b (Table 5). None of 23 pedigreed hatchery-produced age three fish from 2005 were inbred. As a result, the presence of these fish reduced the overall level of inbreeding in the 2008 return and correspondingly increased the N_{eI} estimate to 138 (Table 5).

Discussion

For a population of conservation concern, creation of a multigenerational pedigree enables better estimation of

inbreeding and effective population size than most methods based on coancestry values derived from allele sharing estimation (Pemberton 2008). Moreover, maintenance of a pedigree enables examination of heritability of traits and breeding structure during natural breeding (Szulkin and Sheldon 2008; Grant and Grant 2008) and optimization of parent choice and mating design (Pong-Wong and Williams 2007) in hatchery breeding of the population under study. Development of the two-generation pedigrees for the Cultus Lake sockeye salmon that returned from hatchery spawning in 2004 and for the entire Cultus sockeye breeding population of 2008 enabled a realistic assessment of the amount and sources of genetic variation incorporated into the Cultus Lake sockeye salmon hatchery program and its potential impact in 2008. Maintenance of the pedigree for additional generations will enable assessment of kinship and the conservation value of individuals in the population as well as the reproductive success of both wild- and hatchery-born individuals in natural spawning in Cultus Lake.

The two different methods (ML analysis of allele frequencies and individual parentage assignment) gave similar estimates of the contributions from hatchery- and wild-born parents to their progeny returning from the hatchery spawning of 2004. Parentage assignment allowed the estimation of the number of progeny returning to crosses made at the Cultus Lake lab because there was no significant difference in the allele frequencies of wild-born parents spawned at the two hatchery facilities. Ten of the 14 microsatellite loci employed in the study were highly polymorphic (possessed 10 or more alleles) even in the Cultus Lake population of sockeye salmon. Nevertheless, assignment of some progeny whose parents were both drawn from Families A and/or B to an exclusive pair of parents would not have been possible without use of the hatchery cross records to confirm that a pair of potential parents had actually been mated. For future generations of returning fish, ML estimation of parental contributions may not be possible because the more diverse groups of hatchery-born fish spawned at RCH since 2005 have not been characterized by such highly distinctive allele frequencies. On the other hand, parentage assignment for individual fish will be simplified by a vast reduction in the ‘Aunt and Uncle effect’ (Olsen et al. 2001) resulting from the limited numbers of eggs per family transferred each year from Cultus to RCH after 2001.

The pedigree captured known relationships among hatchery-born sockeye salmon transferred from spawning at Cultus to RCH for rearing, as well as inferred sibling relationships among wild-born fish spawned at both facilities. Most of the wild-born fish bred in 2004 were age four fish from natural spawning in 2000, a year in which 1227 sockeye salmon had entered Cultus Lake to spawn. Over

700 fish born in the wild in 2000, primarily from the wild smolt collection of 2002, were available for estimation of full- and half-sib family structure. Recognition of the sibling relationships among the subset of these fish that was incorporated into the population by hatchery spawning prevented a significant overestimation of their contribution to genetic variation. The alternative of considering all 739 fish as unrelated would have required a parental population size of 1478, a number greater than the total return in the year 2000. Treating related individuals of unknown origin as founders (i.e. assuming that they were unrelated to each other and to other individuals in the pedigree) without correction could have potentially serious consequences for future conservation decisions based on kinship values (Oliehoek and Bijma 2009).

ML estimation of single-generation family structure performed as well as pairwise allele-sharing methods in another sockeye salmon breeding program (Kozfkay et al. 2008) when tested on known samples in which family size was limited to two. Moreover, larger sample sizes and larger sibships confer greater exclusion power to group ML methods and give them an advantage over pairwise methods in family identification (Almudevar and Field 1999; Wang 2004). This increased power improves identification of siblings that share relatively few alleles by chance and of unrelated individuals that share more alleles than expected by chance. Nevertheless, error in the sibships defined for the wild-born G_1 fish lacking parental genotype information undoubtedly occurred. However, in the absence of these somewhat arbitrary sibships, the wild-born G_1 fish would all have been founders in the pedigree and considered unrelated by definition. This assumption would have led to inflated estimates of N_e in G_2 and subsequent generations.

The pedigree has also been established to enable ongoing tracking of relationships in the Cultus sockeye population. Fish spawned at both facilities in years since 2004 have been related not only to each other but also to fish spawned in earlier and later years because of variation in sockeye salmon age of maturity. Moreover, the combined result of captive rearing and the difference in the predominant age of maturity between the natural (age four) and hatchery (age three) environments is that the fish released from supplementation at ICH in a given year are aunts and uncles to many of those released from captive rearing 3 years later. The pedigree established for the population, although by necessity an incomplete one, will enable longer-term assessment of genetic variation in the population as the consequences of small population size and captive breeding on coancestry, inbreeding levels and effective population size are realized.

Incomplete sampling of the adult returns to Cultus Lake each year and the presence of both hatchery- and wild-born

fish in both the hatchery and natural environments since 2007 mean that not all parental fish will be identified in a long-term pedigree. Use of an incomplete pedigree for estimation of inbreeding and effective population size results in underestimation of inbreeding and overestimation of the N_e values (Lutaaya et al. 1999; Ewing et al. 2008). Parameters based on the probability of gene origins, such as the effective number of ancestors or FGE generally depend on shorter segments of the complete pedigree and therefore are less impacted by incomplete information (Boichard et al. 1997).

The large difference between the total and effective number of ancestors for both G_1 and G_2 Cultus sockeye salmon revealed that contributions of the ancestors were highly unequal (Boichard et al. 1997). This disparity was confirmed by the fact that only the 20 most influential ancestors out of a total of 142 were required to account for 50% of genes in G_2 . This imbalance largely resulted from the bottleneck imposed in the pedigree by the disproportionate representation of the small number of families created at Cultus in 2001. Not surprisingly, 11 of the 20 most influential ancestors were members of that lineage. Calculation of the number of ancestors for a population takes into account the loss of genetic variation that has occurred since founding due to bottlenecks, but not the loss due to segregation of parental alleles (Boichard et al. 1997), whereas all sources of gene loss are accounted for in estimation of the FGE for a population (Lacy 1989). Unlike N_e , which decreases over time only if inbreeding or the variance in allele frequencies increases, FGE for a population declines over time due to the accumulating segregational loss of alleles each generation (Boichard et al. 1997). The FGE value of 32 for the Cultus sockeye G_2 indicates that the equivalent of 32 complete founding genomes were still present. The similarity of this value to the N_{eV} estimate of 35 for G_1 suggests that, in fact, the greatest loss of diversity in the pedigree had arisen from the bottleneck rather from the segregational loss of alleles. This result reflects the shallowness of the pedigree (i.e. segregation had occurred only twice) and the highly variable contributions of founding families, both wild- and hatchery-born.

The captive breeding component of the hatchery program began producing large numbers of fish from spawning in 2004, a year in which only 53 wild fish spawned in Cultus Lake. Even if no fish had been removed from the return for hatchery spawning, only 90 sockeye salmon would have spawned in Cultus Lake. The number of effective breeders in a single yearclass of Pacific salmonids (N_b) is generally not greater than one-third of the adult census count (Waples 2004). Therefore, in the absence of the hatchery program, N_b for Cultus sockeye in 2008 would have been unlikely to exceed the value of 54 estimated for

the hatchery-produced G_2 . However, the reproductive potential of the hatchery-produced fish under natural spawning conditions has not yet been assessed. Moreover, even in the absence of any fitness differential between hatchery- and wild-born fish, combined hatchery and natural rearing in a salmonid population generally is expected to reduce N_e over time because of the increased the variation in family size (i.e., successful hatcheries produce larger families than would be produced by natural freshwater rearing). A long-term reduction in N_e due to hatchery production will be avoided in Cultus sockeye salmon only if the hatchery-produced fish contribute to an increase in population size that is sustained after hatchery production ceases (Waples and Do 1994).

The G_1 fish spawned at Rosewall in 2004 were not typical of later captive broodstocks. The high rate of maturation at age three in a small set of families created at Cultus in 2001 likely had both genetic and environmental components, with maturation accelerated both by a high proportion of age three G_0 parents and faster growth in captivity. Over 50% of G_1 parents used at Rosewall and Cultus came from two families reared at Rosewall, each comprising the progeny of a single female mated with two different males. In subsequent years, the hatchery-produced broodstock at Rosewall has been derived from larger numbers of Cultus Lake parents, whose contributions in captive rearing have been largely equalized.

Analyses of the age four G_2 fish confirmed the success of measures implemented to increase genetic diversity in the captive broodstock beyond that represented in the limited crosses made at Cultus in 2001. The substantial genetic impact of G_1 parents derived from the wild smolt collections was primarily responsible for the normalization of allele frequencies and levels of genetic diversity in these fish. In addition, assignment methods indicated that about 10% of the age four G_2 were progeny of Cultus crosses made with wild rather than captively-reared fish, although these crosses contributed less than 5% of the juveniles released. Thus, fish from the supplementation program tended to either mature a year later or survive better than those from the captive breeding program, or both. Unfortunately, differences in the rates of survival cannot be interpreted simply in terms of fitness because fish from the two hatchery sources were distributed unequally among release groups that were treated quite differently and subjected to different levels of predation during freshwater residence in Cultus Lake.

Inbreeding, which may cause loss of fitness in wild populations (Frankham 2010), is of particular concern in re-introduced populations (Biebach and Keller 2010). Any ability to manage the breeding structure of a threatened population that was afforded by captivity is likely lost upon re-introduction to the wild and inbreeding may accumulate

rapidly while population size remains small (Ewing et al. 2008). In the case of salmonids, little is known about the extent to which, if any, individuals mating in the wild engage in inbreeding avoidance during mate selection (Hendry et al. 2004). Captive individuals from the Cultus program were released into an extant population of recent coancestry so that inbreeding may have occurred whether hatchery-produced individuals bred with each other or with wild-born individuals. Additionally, environmental or genetic factors other than inbreeding that reduce the reproductive capacity of hatchery-born animals in the wild (Araki et al. 2007a; Schroder et al. 2008) may impact 'hybrid' progeny resulting from the mating of hatchery- and wild-born parents, leading to a further decline in population productivity.

The 2008 return of 521 sockeye salmon was five times larger than that of 2004, with 40 fish likely produced from wild spawning. This can be viewed as a successful outcome of the captive breeding program although the stated objectives of the conservation measures have not yet been achieved (Bradford et al. 2011). The next significant genetic assessment of the hatchery-produced Cultus Lake sockeye salmon will involve their ability to spawn in the wild and produce progeny that themselves survive to spawn in the wild. A series of studies examining the spawning success of steelhead trout from a supplementation hatchery has produced variable results in which the reproductive capacity in the wild of hatchery-produced trout was estimated to decline by as much as 30% for each generation of hatchery rearing (Araki et al. 2007a, b), a result that modelling indicated could be the result of domestication selection acting upon multiple traits (Araki et al. 2008). Assessment of the next generation indicated that the reduction in reproductive capacity extended to the wild-born fish that were produced by captive fish that spawned in the wild (Araki et al. 2009). Demographic and genetic analysis of sockeye salmon smolt production and adult returns at Cultus Lake over the coming years will enable evaluation of the reproductive capacity of hatchery-produced fish and perhaps reveal the ultimate consequences of the hatchery program on population persistence.

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